A FRAMEWORK FROM AD 599 FALL 26 A1

In-Class Labs/Activity

Keep your learning objectives and discussion posts. Add one activity that generates an artifact.

Sree Kumar Valath · AD 599 · October 2026

THE IDEA

What you already have, and what you add

ALREADY HAVE

Learning objectives

No change. The activity practices an objective you already teach.

ALREADY HAVE

Lecture

No content changes. Make room for one activity segment.

NEW

Activity and artifact

Students build one artifact: a code snippet, document, table or diagram.

ALREADY HAVE

Discussion post

Link the prompt to the activity. Optional: ask students to attach the artifact.

FOR STUDENTS

Learn, comprehend, demonstrate, apply

During the lecture	Discussion board	Beyond the module

Learn

Concepts through the lecture content.

Comprehend

Try it out in the activity and generate an artifact.

Demonstrate

Submit the artifact in the Discussion board and compare with the class.

Apply

In future lessons, assignments, the term project and the real world.

The artifact is what connects the lecture to the discussion: students leave class with what the post needs.

FOR INSTRUCTORS

Why it is easy to integrate

Nothing new to design

No new learning objectives, assignments or grade categories.

No new submission

The artifact reinforces the discussion post students already submit.

Any level of technology

A web app, a form tool, or paper.

Works online

Every step can be followed from the recording if students miss the live session.

TAKING PART

Three ways to take part, one artifact

	Live in class	From the recording	Offline
When	During the activity segment of the lecture	Pause at the activity slide	Any time, with a printed sheet
Input	Unique per student	The same unique input	A versioned sheet or their own data
Submit	Artifact with the discussion post	Same	Same (a photo of the sheet is fine)

A student who misses the live session does the same activity from the recording. Only the timing changes.

ACTIVE EXPERIMENT, FALL 2026

How I run it: a Python web app

Live site: met-ad-599.github.io · Template: github.com/BostonAnalytics/class-artifacts

- Students log in from Day 1, start with a survey, then get a unique question set, data set or team activity each week.
- Each student sees only their own submission; I show class results live from an admin dashboard.
- The class’s own answers become the dataset for later lectures.
- The same link works live and from the recording.

The app is a convenience, not a requirement. Every activity also runs on paper or with a free form tool.

ANY COURSE

No web app? Simple techniques

Think, pair, share

Answer alone, compare with a partner, then share with the room.

Artifact: the written answer

Paper activity

Work through a printed sheet or case by hand.

Artifact: a photo of the sheet

Role play

Act out a real scenario in assigned roles.

Artifact: a short write-up of the decision

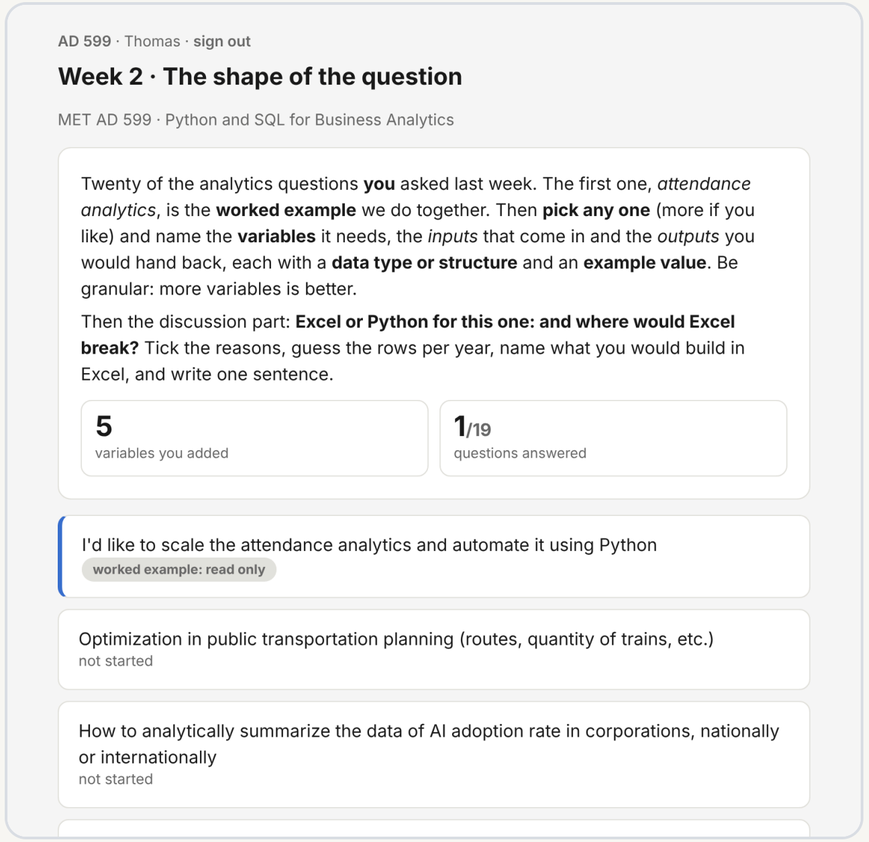
SAMPLE

AD 599: six modules, six artifacts

Module	Activity	Artifact
1. Python and data structures	Data Passport	Data Passport
2. Logic, strings, functions	Rules Engine	Function plus trace table
3. Algorithms and data science packages	Sort It Out	Step count table plus a recommendation
4. Machine learning and loss functions	Human Regression	ML Evaluation Scorecard
5. SQL basics, joins, grouping	Paper Database	Query log
6. Advanced SQL and Python integration	Leaderboard to Pipeline	Pipeline sketch: ERD and system design

/W02/ WALKTHROUGH · STEP 1 OF 3

Students start from their own questions



Their questions

The list is the analytics questions the class asked in the Week 1 survey.

One worked example

Attendance analytics is done together on the slides, read only.

Then their turn

Each student picks a question and builds the artifact for it.

Screens from a test copy of the app with sample data.

/W02/ WALKTHROUGH · STEP 2 OF 3

The artifact: variables plus a verdict

Discussion 2: how the logic works

Your variables

INPUTS

variable	type / structure	example	
parcels	list of dictionaries	`[{"id": "P1042", "zone": "Back Bay", "kg": 2.5}, ...]` one row per parcel	remove
drivers	dictionary	`{"Ann": {"zone": "Back Bay", "capacity_kg": 300}, ...}` look up a driver by name	remove

OUTPUTS

variable	type / structure	example	
assignments	dictionary of lists	`{"Ann": ["P1042", "P1043"], "Josh": ["P1050"]}` each driver gets a list of parcels	remove
load_kg	dictionary	`{"Ann": 284.5, "Josh": 190.0}`	remove
unassigned	list	`["P1099"]` if no van in the zone has room	remove

Discussion 2: why Python is suitable

To solve this problem saved

1. Which tool would you reach for?

Excel is fine for this | **Python** | Either: depends on scale

2. Where would Excel break? (tick any)

- [x] Too many rows for a sheet
- [ ] Needs 3+ tables joined
- [ ] Needs a model or an optimizer, not a formula
- [ ] Needs cleaning before it can be counted
- [x] Has to run every week without me
- [ ] Data is not a flat grid (nested / hierarchical)
- [ ] Comes from an API or many files
- [x] Someone else must be able to re-run it exactly

3. Rows per year, roughly (a sheet holds 1,048,576)

~1 million

4. What would you build in Excel?

Copy-paste + formulas

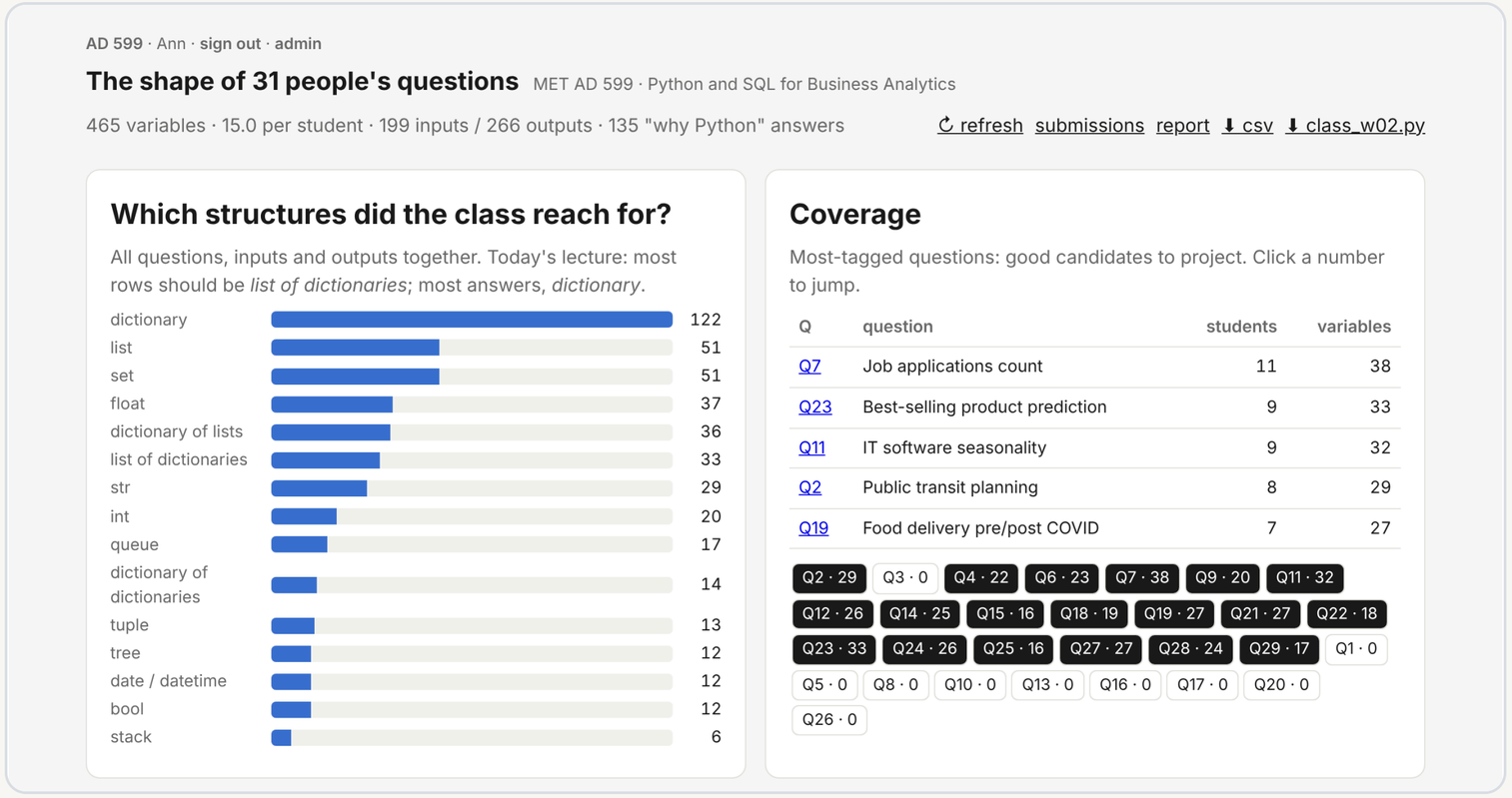
5. Comments

Every morning someone copy-pastes 2,500 parcels and checks van weights by eye. A loop over the parcels with an if/else on capacity does it the same way every day, in seconds.

Update

/W02/ WALKTHROUGH · STEP 3 OF 3

I show the class results live



Private for students

Each student sees only their own work.

Shared in class

I project the class view to debrief.

Sample data.

EXAMPLE FROM THE WEB APP

How /w02/ feeds Discussion 2

met-ad-599.github.io/w02/ · The shape of the question

THE ACTIVITY

Worked example

“I’d like to scale the attendance analytics and automate it using Python.”

Students name the inputs and outputs, then judge Excel or Python and tick where Excel breaks.

DISCUSSION 2 ASKS → THE ARTIFACT HAS

A business scenario to automate
Their own question from the Week 1 survey

How the if/else and loop logic works
Inputs and outputs, typed: scan_log in, at_risk out (rate below 70%)

Why Python is a suitable tool
Too many rows, runs every week, must re-run exactly; copy-paste in Excel becomes a for loop

NEXT STEP

Getting started: one module at a time

1. Pick one of the module’s existing learning objectives.
2. Choose the artifact students will hand in.
3. Place the activity right after the lecture segment whenever a Learning Objective is taught.
4. Prepare unique inputs: a web app, versioned sheets, or the student’s own data. I do a combination of all depending on the lecture.
5. [Especially for ONLINE] Link the discussion prompt to the activity and ask for the artifact. So the grading rubric must account for inclusion of class activity in the post.